



FUJIAN JIANGLE DAYAN 32MW HYDROPOWER PROJECT

Document Prepared by Beijing 3050 Technology Co., Ltd

Project title	Fujian Jiangle Dayan 32MW Hydropower Project
Project ID	246
Monitoring period	28-March-2006 to 27-March-2012
Original date of issue	25-April-2024
Most recent date of issue	25-July-2024
Version	02
VCS Standard Version	VCS-Standard-v4.7
Prepared by	Beijing 3050 Technology Co., Ltd

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PROJECT DETAILS

1.1 Summary Description of the Implementation Status of the Project

Fujian Jiangle Dayan 32MW Hydropower Project (hereafter referred to the project) is located on Jinxi River, the tributary of Min River, 55 km away from Jiangle County seat. It is the third-grade station of Jinxi River cascade hydropower development, with total installed capacity of 32 MW. The Project proponent is Fujian Jinhu Power Co., Ltd.

The project is a low water head run-of-river hydropower plant, and is the third-grade project. The surface area is 1.871 km² at full reservoir level, and the reservoir's Power Density is about 17.10 W/m². The total electricity produced by the project activity (TEG_y) is 121,990 MWh, the annual electricity export to the East China Power Grid (ECPG) through Fujian Grid from the project is estimated to be 120,282 MWh.

The project will reduce the import and increase the export of the electricity to ECPG, subsequently displace power generation by coal-fired power plants, which is the dominant power generation of ECPG. Therefore, the project will obtain greenhouse gas, GHG, emission reduction by avoiding CO₂ emission produced from the business-as-usual scenario in Fujian province. The estimated average annual GHG emission reductions during the first crediting period will be 105,990 tCO₂e.

The construction of the project was started on 04-April-2003, 2 units of 16MW turbine and generator sets were commissioned on 15-December-2004, 25-February-2005 respectively.

The project has operated without any accidental or emergency events that might affect the accuracy and/or implementation of the monitoring activities. The net power supply during this monitoring period (28-March-2006 to 27-March-2012) is 634,403.386MWh. The total emission reductions claimed in this monitoring period are 563,855tCO₂e.

1.2 Audit History

Audit type	Period	Program	Validation/verification body name	Number of years
Validation	Registration	VCS	<u>TÜV SÜD INDUSTRIE SERVICES GMBH</u>	/
Verification	28-March-2006 to 27-March-2012	VCS	<u>LGAI Technological Centre, S. A. (Applus+ Certification)</u>	6 years
Total	-	-	-	6

1.3 Sectoral Scope and Project Type

Sectoral scope¹	1: Energy industries (renewable /non-renewable sources);
Project activity type	Renewable energy project;

1.4 Project Proponent

Organization name	Fujian Jinhu Power Co., Ltd.
Contact person	Xinghua Xiao
Title	Director
Address	Jiangle County, Sanming City Fujian Province, People's Republic of China
Telephone	+86-0598-2358868
Email	40399942@qq.com

1.5 Other Entities Involved in the Project

Organization name	Beijing 3050 Technology Co.,Ltd
Role in the project	Consultant
Contact person	Jiangzhuo Zheng
Title	Project Manager
Address	Room 80925, 4 th floor, building 3, 116 Xinhua East Street, Tongzhou District, Beijing PR China
Telephone	+86 18010126150
Email	jiangzhuo@agriforestry.net

¹ Projects, activities, or methodologies may be developed under any of the 16 VCS sectoral scopes: <https://verra.org/programs/verified-carbon-standard/vcs-program-details/#sectoral-scopes>

1.6 Project Start Date

Project start date	15-December-2004
Justification	The date is the commissioning date of the first turbine generator, which meets the requirement of VCS 2007 (The project start date for the VCS 2007 shall not commence before 01-January-2002).

1.7 Project Crediting Period

Crediting period	Other:10-year renewable (renewable twice) (The crediting period as specific under the VCS 2007 and the earliest starting date of the crediting period is 28-March-2006 as per VCS 2007)
Start and end date of first or fixed crediting period	28-March-2006 to 27-March-2016

1.8 Project Location

The project is located at the upper reach of Jinxi River Basin, 300 m upstream away from Dayan Village, Huangtan Town, Jiangle County, Fujian province, P.R. China.

Geographical coordinates of the project's Gravity Dam and the powerhouse is 26.6543° N, 117.2575° E.

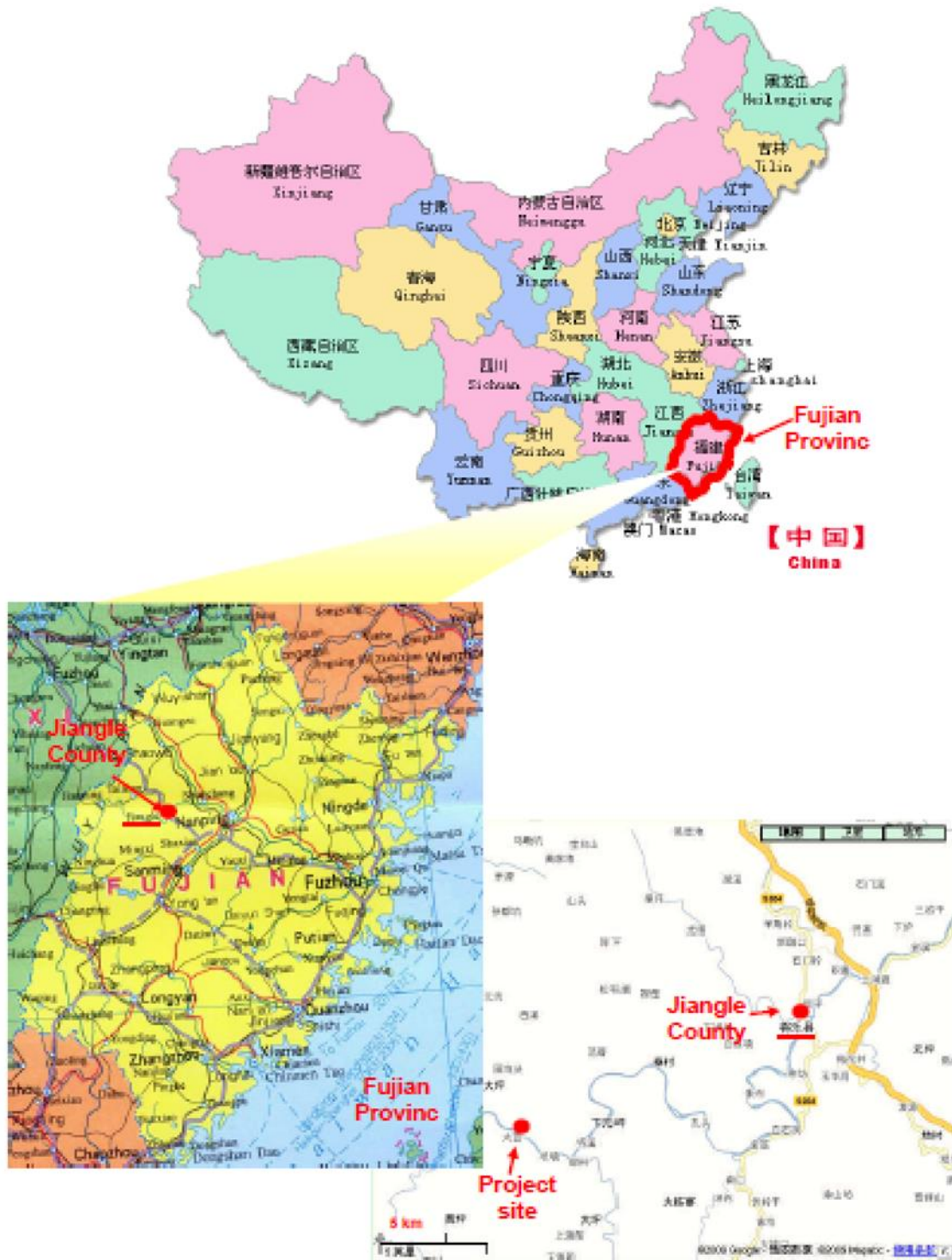


Figure 1 the location of the project site

1.9 Title and Reference of Methodology

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	ACM0002	“Consolidated baseline methodology for grid-connected electricity generation from renewable sources”	10.0
Tool	-	Tool for the Demonstration and Assessment of Additionality	05.2
Tool	-	Tool to calculate the emission factor for an electricity system	01.1

1.10 Double Counting and Participation under Other GHG Programs

1.10.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

1.10.2 Registration in Other GHG Programs

Is the project registered or seeking registration under any other GHG programs?

☐ Yes ☒ No

1.11 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.11.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

China launched a national emissions trading scheme at 16-July-2021 only covering the fossil fuel fired power industry. Other regional mandatory pilot emissions trading schemes which operated earlier in China include carbon intensive industries such as thermal power generation, petrochemical, chemical, building materials, iron and steel, non-ferrous, paper, aviation etc., not including renewable energy sector , The project activity is not included in any Chinese mandatory

emissions trading scheme and there is no emission cap enforced for the Project proponent according to the compliance company list publicly available.

As checked with Clean Development Mechanism (CDM), I-REC, GCC and GS websites, the project was not registered under CDM, I-REC, GCC and GS.

1.11.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the VCS Program Definitions for definition of GHG-related environmental credit system.

☐ Yes ☒ No

1.11.3 Supply Chain (Scope 3) Emissions

Do the project activities affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

1.12 Sustainable Development Contributions

The project is located at the upper reach of Jinxi River Basin, 300 m upstream away from Dayan Village, Huangtan Town, Jiangle County, Fujian province, P.R. China, involves to employ 34 employees, and operates of 32 MW(2*16MW) hydro-generator units and generate power from clean renewable hydro power in Fujian Province.

- The project has supplied 634,403.386 MWh in this monitoring period that otherwise have been produced by thermal power generation projects, and contributes to Chinese renewable energy target.
- The project generated 34 decent working opportunities for local women and men, 17 for women and 17 for men and help them improve the incomes, and contributes to the decent work and economic growth.
- The emission reduction of the project is 563,855 tCO₂e in this monitoring period, and contributes to Chinese Climate change target.

Table 1: Sustainable Development Contributions

Row number	SDG target	SDG indicator	Net impact on SDG indicator	Current project contributions	Contributions over project lifetime
1)	7.2	7.2.1 Renewable energy share in the total final energy consumption	Implemented activities to increase	The project utilizes hydro energy for electricity generation, which could increase the ratio of renewable energy in total energy consumption of China. From 28-March-2006 to 27-March-2012 the electricity supplied from the project is 634,403.386MWh.	From the operation start date of the project to the end of this monitoring period, the electricity supplied from the project is around 634,403.386 MWh.
2)	8.5	Achieve full and productive employment and decent work for all women and men	Implemented activities to increase	During this monitoring period (28-March-2006 to 27-March-2012), the project achieves full and productive employment and decent work for all women and men. In 2006, there are 34 jobs created for 17 men and 17 women, the average yearly salary is 32,996.93RMB/year In 2012 there are 34 jobs created for 17 men and 17 women, the average yearly salary is 41,905.08RMB/year	From the start date of the crediting period of the project (28-March-2006) to the end of this monitoring period, the project provided 34 jobs for women and men from the year of 2006 to the year of 2012, women, and men both are worked for the project, and the average yearly salary has increased from 32996.93RMB/year in 2006 to 41,905.08RMB/year in 2012. The project achieves full and productive employment and decent work for all women and men.

3)	13.0	Tonnes of greenhouse gas emissions avoided or removed	Implemented activities to increase	By utilizing water for power generation and replaced it from thermal power generation projects From 28-March-2006 to 27-March-2012 the emission reduction from the project is around 563,855 tCO₂e	From the operation start date of the project to the end of this monitoring period, the emission reduction from the project is around 563,855 tCO₂e.
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1.13 Commercially Sensitive Information

No commercially sensitive information.

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

Stakeholder Identification

The following steps were adapted to identify stakeholders.

Step 1: Project Documents Review

Stakeholder Identification was started by reviewing project documents. E.g. project approval, feasibility study report (FSR), environmental impact assessment (EIA), contract documents etc. These documents not only outline the primary objectives and the intended outcomes of the project, but also mention stakeholders impacted by the project and provide initial insights into who the key stakeholders might be. After the documents review, the project team listed down the potential stakeholders impacted by the project through discussion.

Step 2: local stakeholders Survey

Before the local stakeholder's survey was carried out, the project team issues a stakeholder solicitation announcement and the stakeholders can contact the project team through telephone and other means specified in the announcement to express their expectations, concerns and needs. The project team communicates with local residents around the project area to collect their opinions. Their perspective helps identify concerns and potential stakeholders that may not be immediately obvious.

Step 3: Regulatory and Legal Research

	The project team researched the local and national regulations and laws that pertain to the hydropower projects. This is helpful to identify regulatory bodies, agencies, and government representatives who may be stakeholders due to their role in permitting, compliance, or oversight. Step 4: Expert Consultations The project team consulted with experts in the field of the hydropower project and environmental impact assessments. They can provide insights into the potential stakeholders who should be considered due to their expertise or interest in the project. Step 5: Stakeholder Catalogue The project team created a stakeholder catalogue. The stakeholders identified: 1: Dayan Villagers 2: Jiangle County government 3: The project staff
Legal or customary tenure/access rights	The project is located at the upper reach of Jinxi River Basin, 300 m upstream away from Dayan Village, Jiangle County, Fujian province, China. As per FSR and its approval, the implementation of the project would result the submerge of 127 Mu cultivated land, 23.5 Mu Forest land and 63 villagers migrated, which is collective owned land and held by local communities, which has been compensated according to relevant local and national regulation, the legal or customary tenure/access rights to territories and resources was protected as per the local and national law/regulations.
Stakeholder diversity and changes over time	Following is the analysis of the social, economic, and cultural diversity within the stakeholder groups involved in the project: 1: Dayan Villagers

Social Diversity: The local residents' group may comprise a mix of age groups, from young families to elderly individuals. They may have different levels of education, occupations, and family sizes.

Economic Diversity: Income levels among this group can vary. Some may rely solely on farming, while others might have additional sources of income or employment in non-agricultural sectors.

Cultural Diversity: The local area is predominantly inhabited by Han Chinese. This group shares similar background and traditions. Therefore, the cultural diversity within this group is negligible.

2: Dayan County government

Within the local government of a County primarily inhabited by Han Chinese, there is generally limited social, economic, and cultural diversity. Government officials typically hail from the local area, sharing similar social backgrounds, economic statuses, and cultural traditions. As such, diversity within the local government is low compared to larger cities or multiethnic regions.

3: The project staff

This project is located in a County, primarily employs individuals from the local communities. The social and cultural diversity of this group (project staff) is largely in line with that of the local residents. In terms of economic diversity, while the group of local resident experiences varying income levels due to diverse sources of income, this group (Project staff) all work within the same project, resulting in similar income levels and thus minimal economic diversity among them.

The diversity mentioned above is unlikely to change over time.

The interactions between these stakeholders are characterized by a mix of cooperation, collaboration, and engagement aimed at addressing shared concerns, achieving common goals, and promoting the well-being of the local community.

Expected changes in well-being	Economic well-being The project has supplied renewable energy to local area, reduce the electricity shortage in the coastal area of Fujian province and contribute to the economic growth in the area. Social well-being The project generated 34 working opportunities for local residents, and help them improve the incomes and the likelihood. Environmental well-being The project has supplied renewable energy to local area that otherwise have been produced by thermal power generation projects, the implementation of the project would reduce the pollutants of SO₂, NO_x, particulate matter and GHG emission. In summary, the project is expected to bring about a range of positive changes, including economic, environmental, and health benefits. These changes are relative to the baseline scenario of producing electricity mainly from fossil fuel fired power plants, which can be resource-intensive and environmentally harmful.
Location of stakeholders	The project is located at the upper reach of Jinxi River Basin, 300 m upstream away from Dayan Village, Jiangle County, Fujian province, China. Which is the third level hydropower projects in Jinxi River, 10KM away from the second level of Liangqian Hydropower project and 16KM away from the fourth level of Chitan hydropower project as well as 55km away from Jiangle County.
Location of resources	The project is located at the upper reach of Jinxi River Basin, 300 m upstream away from Dayan Village, Jiangle County, Fujian province, China. Which is the third level hydropower projects in Jinxi River, 10KM away from the second level of Liangqian Hydropower project and 16KM away from the fourth level of Chitan hydropower project

2.1.2 Stakeholder Consultation and Ongoing Communication

Ongoing consultation	The Project proponent has established a grievance and appeal mechanism, a grievance book is kept at the Project proponent office. Anyone can go there and record their grievance or comment in the book. The Project proponent personnel would review and respond to the comments and record the response in the book as well. Meanwhile, the local authority has also conducted spot checks on the implementation of the project from time to time as per the request from the local governments' regulations.																		
Date(s) of stakeholder consultation	The grievance book was available after the project implementation and constant stakeholder consultations were carried out before the project implementation on 01-September-2004 and after the project implementation from the year of 2005's stakeholder consultation on 06-September-2005 as well as the stakeholder consultations during this monitoring period from 2006's to 2012's. In each stakeholder consultation there are 30 questionnaires were distributed to the stakeholders and all the questionnaires were collected, the date(s) of stakeholder consultation are showed as below <table border="1"> <thead> <tr> <th>Year</th><th>Date(s) of stakeholder consultations</th></tr> </thead> <tbody> <tr> <td>2004</td><td>01- September -2004</td></tr> <tr> <td>2005</td><td>06- September -2005</td></tr> <tr> <td>2006</td><td>09- September -2006</td></tr> <tr> <td>2007</td><td>02- September -2007</td></tr> <tr> <td>2008</td><td>04- September -2008</td></tr> <tr> <td>2009</td><td>05- September -2009</td></tr> <tr> <td>2010</td><td>04- September -2010</td></tr> <tr> <td>2011</td><td>02- September -2011</td></tr> </tbody> </table>	Year	Date(s) of stakeholder consultations	2004	01- September -2004	2005	06- September -2005	2006	09- September -2006	2007	02- September -2007	2008	04- September -2008	2009	05- September -2009	2010	04- September -2010	2011	02- September -2011
Year	Date(s) of stakeholder consultations																		
2004	01- September -2004																		
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2006	09- September -2006																		
2007	02- September -2007																		
2008	04- September -2008																		
2009	05- September -2009																		
2010	04- September -2010																		
2011	02- September -2011																		

	2012	07-September-2012																		
Communication of monitored results	There is no grievance or negative comments from the stakeholders before and during this monitoring period. The questions of 1: Is the project helpful to local economic and social development? (Yes, No, not sure) 2: Does the project influence the aquatic and terrestrial animals and plants? (Yes, No, not sure) 3: Does the project influence the downstream ecology and water supply? (Yes, No, not sure) 4: Do you think the project improve your livelihood (Yes, No, not sure) 5: Do you think the project improve the environment (Yes, No, not sure) 6: Do you support the project (Yes, No, not sure) 7: What do you think about the measures adopted in environment and ecology by the company? (Satisfied, Unsatisfied, not sure) The summary of survey results is showed as below 2004(before the project implementation) Summary of occupations/jobs <table><tr><th>Stakeholders</th><th>Occupations</th><th>Amount of Questionnaires</th><th>Ratio</th></tr><tr><td>Local government</td><td>-</td><td>3</td><td>10.0%</td></tr><tr><td rowspan="3">Dayan Villagers</td><td>Farmers</td><td>8</td><td>26.7%</td></tr><tr><td>workers</td><td>7</td><td>23.3%</td></tr><tr><td>Other residents</td><td>10</td><td>33.3%</td></tr></table>		Stakeholders	Occupations	Amount of Questionnaires	Ratio	Local government	-	3	10.0%	Dayan Villagers	Farmers	8	26.7%	workers	7	23.3%	Other residents	10	33.3%
Stakeholders	Occupations	Amount of Questionnaires	Ratio																	
Local government	-	3	10.0%																	
Dayan Villagers	Farmers	8	26.7%																	
	workers	7	23.3%																	
	Other residents	10	33.3%																	

The project staff	-	2	6.7%
The ratio of male again female			17: 13

The results of the Questionaries

No	Result of the survey
1	93.3% of the participants think the project facilitate to local economic and social development; and no one chooses the negative option.
2	0% of the participants think the project influence the aquatic and terrestrial animals and plants.
3	0% of the participants think the project influence the downstream ecology and water supply.
4	93.3% think improve the likelihood and no one chooses the negative option.
5	100% think improve the environment
6	100% support the project
7	90% are satisfied with the measures adopted in environment and ecology by the company; and no one chooses the negative option.

2005(after the project implementation but before this monitoring period)

Summary of occupations/jobs

	Occupations	Amount of Questionaries	Ratio
Local government	-	4	13.3%
Dayan Villagers	Farmers	9	30.0%
	workers	6	20.0%
	Other residents	9	30.0%

The project staff	-	2	6.7%
The ratio of male again female			16:14

The results of the Questionaries

No	Result of the survey
1	93.3% of the participants think the project facilitate to local economic and social development; and no one chooses the negative option.
2	0% think the project influence the aquatic and terrestrial animals and plants.
3	0% think the project influence the downstream ecology and water supply.
4	93.3% think improve the likelihood and no one chooses the negative option.
5	90% think improve the environment, no one chooses the negative option.
6	100% support the project
7	90% are satisfied with the measures adopted in environment and ecology by the company; and no one chooses the negative option.

2006(during this monitoring period)

Summary of occupations/jobs

	Occupations	Amount of Questionaries	Ratio
Local government	-	2	6.7%
Dayan Villagers	Farmers	6	20.0%
	workers	8	26.7%
	Other residents	11	36.7%

The project staff	-	3	10.0%
The ratio of male again female			17:13

The results of the Questionaries

No	Result of the survey
1	93.3% of the participants think the project facilitate to local economic and social development; and no one chooses the negative option.
2	0% think the project influence the aquatic and terrestrial animals and plants.
3	0% think the project influence the downstream ecology and water supply.
4	93.3% think improve the likelihood and no one chooses the negative option.
5	90% think improve the environment, no one chooses the negative option.
6	100% support the project
7	90% are satisfied with the measures adopted in environment and ecology by the company; and no one chooses the negative option.

2007(during this monitoring period)

Summary of occupations/jobs

	Occupations	Amount of Questionaries	Ratio
Local government	-	3	10.0%
Dayan Villagers	Farmers	8	26.7%
	workers	9	30.0%
	Other residents	7	23.3%

The project staff	-	3	10.0%
The ratio of male again female			18:12

The results of the Questionaries

No	Result of the survey
1	90% of the participants think the project facilitate to local economic and social development; and no one chooses the negative option.
2	0% think the project influence the aquatic and terrestrial animals and plants.
3	0% think the project influence the downstream ecology and water supply.
4	90% think improve the likelihood and no one chooses the negative option.
5	90% think improve the environment, no one chooses the negative option.
6	100% support the project
7	90% are satisfied with the measures adopted in environment and ecology by the company; and no one chooses the negative option.

2008(during this monitoring period)

Summary of occupations/jobs

	Occupations	Amount of Questionaries	Ratio
Local government	-	2	6.7%
Dayan Villagers	Farmers	6	20.0%
	workers	8	26.7%
	Other residents	10	33.3%

The project staff	-	4	13.3%
The ratio of male again female			16:14

The results of the Questionaries

No	Result of the survey
1	93.3% of the participants think the project facilitate to local economic and social development; and no one chooses the negative option.
2	0% think the project influence the aquatic and terrestrial animals and plants.
3	0% think the project influence the downstream ecology and water supply.
4	93.3% think improve the likelihood and no one chooses the negative option.
5	90% think improve the environment, no one chooses the negative option.
6	100% support the project
7	90% are satisfied with the measures adopted in environment and ecology by the company; and no one chooses the negative option.

2009(during this monitoring period)

Summary of occupations/jobs

	Occupations	Amount of Questionaries	Ratio
Local government	-	3	10.0%
Dayan Villagers	Farmers	8	26.7%
	workers	8	26.7%
	Other residents	8	26.7%

The project staff	-	3	10.0%
The ratio of male again female			15:15

The results of the Questionaries

No	Result of the survey
1	93.3% of the participants think the project facilitate to local economic and social development; and no one chooses the negative option.
2	0% think the project influence the aquatic and terrestrial animals and plants.
3	0% think the project influence the downstream ecology and water supply.
4	90% think improve the likelihood and no one chooses the negative option.
5	90% think improve the environment, no one chooses the negative option.
6	100% support the project
7	90% are satisfied with the measures adopted in environment and ecology by the company; and no one chooses the negative option.

2010(during this monitoring period)

Summary of occupations/jobs

	Occupations	Amount of Questionaries	Ratio
Local government	-	3	10.0%
Dayan Villagers	Farmers	8	26.7%
	workers	9	30.0%
	Other residents	7	23.3%

The project staff	-	3	10.0%
The ratio of male again female			17:13

The results of the Questionaries

No	Result of the survey
1	90% of the participants think the project facilitate to local economic and social development; and no one chooses the negative option.
2	0% think the project influence the aquatic and terrestrial animals and plants.
3	0% think the project influence the downstream ecology and water supply.
4	90% think improve the likelihood and no one chooses the negative option.
5	90% think improve the environment, no one chooses the negative option.
6	100% support the project
7	90% are satisfied with the measures adopted in environment and ecology by the company; and no one chooses the negative option.

2011(during this monitoring period)

Summary of occupations/jobs

	Occupations	Amount of Questionaries	Ratio
Local government	-	2	6.7%
Dayan Villagers	Farmers	6	20.0%
	workers	8	26.7%
	Other residents	11	36.7%

The project staff	-	3	10.0%
The ratio of male again female			14:16

The results of the Questionaries

No	Result of the survey
1	90% of the participants think the project facilitate to local economic and social development; and no one chooses the negative option.
2	0% think the project influence the aquatic and terrestrial animals and plants.
3	0% think the project influence the downstream ecology and water supply.
4	93.3% think improve the likelihood and no one chooses the negative option.
5	100% think improve the environment
6	100% support the project
7	90% are satisfied with the measures adopted in environment and ecology by the company; and no one chooses the negative option.

2012(during this monitoring period)

Summary of occupations/jobs

	Occupations	Amount of Questionaries	Ratio
Local government	-	4	13.3%
Dayan Villagers	Farmers	9	30.0%
	workers	6	20.0%
	Other residents	9	30.0%

	The project staff	-	2	6.7%
	The ratio of male again female			16:14
	The results of the Questionaries			
	No	Result of the survey		
	1	93.3% of the participants think the project facilitate to local economic and social development; and no one chooses the negative option.		
	2	0% think the project influence the aquatic and terrestrial animals and plants.		
	3	0% think the project influence the downstream ecology and water supply.		
Consultation records	4	90% think improve the likelihood and no one chooses the negative option.		
	5	90% think improve the environment, no one chooses the negative option.		
	6	100% support the project		
	7	93.3% are satisfied with the measures adopted in environment and ecology by the company; and no one chooses the negative option.		
Stakeholder input		The grievance book and stakeholder consultation records		
		There are no negative comments received for the project during this monitoring period. In line with VCS requirements all the processes have been implemented to receive comments from local stakeholders as well as communicate with them at periodic intervals.		

2.1.3 Free, Prior, and Informed Consent

Consent	The grievance book was kept on the Project proponent office to gain the feedback from the stakeholder, and a stakeholder consultation were carried out on a regular basis and no negative feedback was received during this monitoring period, the project obtained the informed
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	consent to implement through stakeholder consultation before the project implementation by preparing and submitting the feasibility study report, environmental assessment report and other documents to the relevant government departments to gain the local authority approval. The relevant villagers with cultivated and forest land submerged and the villagers migrated had gain the compensation as per the local and national regulations as per the signed contracts provided, all of them agreed on the compensation, meanwhile all of the stakeholders including the villagers directly impacted by the project are supposed the implementation of the projects, as per the result of the grievance book and the stakeholder consultation, specially the villagers directly impacted by the projects does not have unresolved conflicts happened during the project implementation, which means the project resolved all the conflicts before the project implementation. The project is contributing to the sustainable development from the aspect of social, economic and environmental, therefore the project does not exacerbate nor influence the outcomes of unresolved conflicts. There are no ongoing or unresolved conflicts.
Outcome of FPIC	The project obtained the Project Approval issued by Fujian Development and Reform Commission,04-June-2002. The project obtained the EIA approval issued by Fujian Environmental Protection Bureau on 12-February-1999. The relevant stakeholders with the submerge of 127 Mu cultivated land, 23.5 Mu Forest land and 63 villagers are migrated, all are collective owned land and held by local communities, have gained the compensation according to relevant local and national regulation, the legal or customary tenure/access rights to territories and resources was protected as per the local and national law/regulations, they are all agreed the implementation of the project. As per the signed contracts provided, the stakeholders with submerged cultivated land/forest land and migrated

villagers are all agreed with the compensations as per the local and national regulations, as well as the implementation of the project, therefore the project does not involve in the situation of the encroached-on land, relocated people without consent, and forced physical or economic displacement.

2.1.4 Grievance Redress Procedure

No grievance was received during this monitoring period.

Grievances received	Resolution and outcome
Summarize the grievance raised during the monitoring period.	There is no grievance raised during this monitoring period as per the grievance book used for collecting the comments and feedback from the stakeholders. As the Project proponent office is 300 meters upstream away from Dayan Village, where the main stakeholders are located and easy for them to get access to the grievance book to express their opinions and comments regarding the project.

2.1.5 Public Comments

Summary of comments received	Actions taken
The project was webhosted for public comments, and <i>no public comments received</i>	Not applicable

2.2 Risks to Stakeholders and the Environment

	Risk identified	Mitigation or preventative measure taken
Risks to stakeholder participation	No risks identified	Not applicable. The implementation of the project has reduced the emission of SO₂, NO_x etc., and the stakeholders involved in the project are very supportive of the project.
Working conditions	No risks identified	Not applicable

		Most of time, the staff work in indoor. In case of working outside, the Project proponent provides safety protection equipment and safety measures to ensure their safety.
Safety of women and girls	No risks identified	Not applicable China has published “Women’s Rights Protection Law ² ” to provide support and protection to women and girls, the women/girls staff work for the project are well protection as demonstrated in item of “working conditions”.
Safety of minority and marginalized groups, including children	No risks identified	Not applicable The project does not involve in minority and marginalized groups including Children. China has published “Protection Laws on the Juvenile ³ ”. Employment regulations are described in this Law. The Project requires skilled employees to operate, maintain, and manage the facilities/equipment’s. Therefore, the project does not employ and is not complicit in any form of child labor. According to the Employee list, it could be confirmed that the Project does not employ any child labor.
Pollutants (air, noise, discharges to water, generation of waste, release of hazardous materials)	No risks identified	Not applicable According to the EIA report of the project, environmental impacts caused by the project and corresponding measures adopted by the Project proponent are illustrated as the following: 1: Impact on social economics The loss of submerged properties of the project is small. The natural condition of the Dayan Project is an advantage for its construction; the economic benefit is quite significant. The project will contribute to rural electrification development and to alleviate electricity shortage in Jiangle County. It is an ideal site for construction of hydropower station. Building up of the project complies with the Jinxi River cascade hydropower station development, which is to promote the development of hydro powers. The project also meets the requirement of The Comprehensive Plan and Design of Jinxi River, which is a guideline from local government to mark out the development of the basin. 2: Impact on quality of surface water

² https://www.gov.cn/guoqing/2021-10/29/content_5647634.htm

³ https://www.gov.cn/xinwen/2020-10/18/content_5552113.htm

Waste water discharged during the construction period mostly comes from construction process and daily life. The construction company will construct deposit pool, sedimentation pool and oil blocking pool to dispose the waste water from construction; cesspool will be established to mitigate the direct pollution of waste water from daily life.

3: Eutrophication impact

Based on the research of reservoir eutrophication impact both in the country and overseas, Total Phosphorous (P) is selected as the assessment indicator of eutrophication impact.

Phosphorous was not detected before the reservoir construction. Phosphorous may come from daily life, soil erosion and emerged forest land, plantation and plants. Using an assessment model (Dillon Model) to calculate the P content in the reservoir after the project begin operation, the result is 0.013mg/L. According to the standard of P level listed below, Dayan reservoir will remain poor nutrition so that eutrophication will not happen.

Level	Poor	media	Eutrophication
P (mg/m ³)	<50	50-100	>100

4: Environmental impact during construction

The troubling and organic matters are increased in the lower reach of the river, the noise standard of living and office area is higher than the normal. There are dust produced in the construction area, however, the impact can be reduced by installing the dust shields for the operation machines. The Project proponent shall also plant trees and grasses around the project site to reduce noise and dust.

5: Impact on soil erosion

The project occupies land 127 mu, includes forest and plantation. The soil erosion affected area includes 23.5 mu forest, 114 mu plantation.

There is solid waste from the project construction, vegetation recovery and proper handling the solid waste are primary concern for preventing the soil erosion.

In conclusion of the EIA report, the project does not have any negative significant impacts on the environment.

2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

Discrimination and sexual harassment	There is no discrimination and sexual harassment happened during this monitoring period. China has published “Labor Law” ⁴ and “Women’s Rights Protection Law” ⁵ , which are the basis of the laws for prevent such kind of phenomena in the workplace. The project abides by the rules of equality and does not involve in any form of discrimination and sexual harassment against the laws.
Management experience	The project proponent is a State-owned enterprise (SEO), and operated many hydropower projects, who has rich management experience in this regard.
Gender equity in labor and work	Equal opportunities have been provided in the context of gender equity and equal pay for labor and work, the project with 34 staff in total with half of staff are female.
Human trafficking, forced labor, and child labor	China has published “Labor Law” and “Protection Laws on the Juvenile” ⁵ . The project is strictly followed the laws and does not involve in human trafficking, forced labor, and child labor.

2.3.2 Human Rights

The project respects internationally proclaimed human rights. China has its own legislation in place prohibiting the violation of human rights principle and it actively enforces the compliance of such principle. China has ratified two UN human rights treaties--the International Covenant on Civil and Political Rights, the International Covenant on Economic, Social, and Cultural Rights. China has published relevant regulations and practice into line with the treaties, such as “Labor Law”.

This project does not discriminate with regards to participation and inclusion. The project abides the rules of equality accordingly and does not involve and is not complicit in any form of discrimination. Specific to gender equality, detailed enforcement rules are regulated in “Women’s Rights Protection Law” to provide support and benefits to women.

⁴ https://www.gov.cn/banshi/2005-05/25/content_905.htm

⁵ http://www.npc.gov.cn/npc/c2/c30834/202210/t20221030_320091.html

2.3.3 Indigenous Peoples and Cultural Heritage

There is no cultural heritage or indigenous Peoples at or near the project site as per the EIA report.

2.3.4 Property Rights

Disputes over rights to territories and resources	N/A. There are no disputes over rights to territories and resources (The relevant stakeholders with the submerge of 127 Mu cultivated land, 23.5 Mu Forest land and 63 villagers are migrated, all are collective owned land and held by local communities, and has gained the compensation according to relevant local and national regulation, the legal or customary tenure/access rights to territories and resources was protected as per the local and national law/regulations)
Respect for property rights	Not applicable The relevant stakeholders involved in the submerge of cultivated land and forest land have gained the compensation before the project implementation. In the implementation stage it does not refer to “Respect for property rights”

2.3.5 Benefit Sharing

Summary of the benefit sharing plan	N/A The relevant stakeholders involved in the submerge of cultivated land and forest land were gained the compensation before the project implementation as per the local and national compensation laws, and all the stakeholders agreed on the implementation of the project.
Benefit sharing during the monitoring period	N/A The relevant stakeholders involved in the submerge of cultivated land and forest land were gained the compensation before the project implementation as per the local and national compensation laws.

2.4 Ecosystem Health

	Risk identified	Mitigation or preventative measure taken during the monitoring period
Impacts on biodiversity and ecosystems	No risk identified	The project proponent has strictly followed the result of EIA report regarding biodiversity and ecosystems eg a minimum ecological waterflow requested, a regular inspection was carried out to ensure its implementation As per public source from Ministry of Ecological and Environment(MEE), Jinxi river Jiangle segment where the project located was awarded as the second batch of excellent cases of beautiful rivers and lakes by MEE ⁶ .
Soil degradation and soil erosion	No risk identified	Not applicable Planting new trees and grass along the two sides of the newly built reservoir and protect the forest land by avoid harvesting tree and reserve the water and soil along the reservoir to avoid soil degradation and soil erosion.
Water consumption and stress	No risk identified	Not applicable The implementation of the project improves the water resource management capacities, which do good to the local communities for daily and agricultural water usage, rather than increase the water consumption and stress.
Usage of fertilizers	No risk identified	Not applicable The implementation of the project does not involve in the usage of fertilizers during this monitoring period, and there is no risk identified.

2.4.1 Rare, Threatened, and Endangered species

Species or habitat	The project located in did not involve in habitats for rare, threatened, or endangered species as per the EIA report.
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2.4.2 Introduction of species

As per the EIA report, the area that the project located does not involve in the rare, threatened and endangered species.

Species introduced	Classification	Justification for use	Adverse effects and mitigation
Not applicable	Not applicable	Not applicable	Not applicable

⁶ <https://mp.weixin.qq.com/s/DrJC0hBmrAljpNwYRMUCbQ>

Existing invasive species	Mitigation measures to prevent spread or continued existence of invasive species
No existing invasive species	Not applicable

2.4.3 Ecosystem conversion

The project is a hydropower project, not ARR, ALM, WRC or ACoGS projects, therefore it is not applicable.

3 IMPLEMENTATION STATUS

3.1 Implementation Status of the Project Activity

The project with total installed capacity of 32MW(2*16MW). The purpose of the project is to deliver electricity to East China Power Grid (ECPG), displacing electricity generated by thermal power plants in ECPG, thus GHG emission reduction is achieved.

The construction of the project was started on 04-April-2003, 2 units of 16MW turbine and generator sets were commissioned on 15-December-2004, 25-February-2005 respectively.

The technical specifications of the turbine/generator units of the project are listed in Table 2: The key parameter of the equipment.

Table 2: The key parameter of the equipment

Turbine(s)		Generator(s)	
Number of Units	2	Number of Units	2
Turbine Type	GZD278A-WP-470	Type	SFWG16-52/1340
Rated capacity	16.667	Rated power	16MW
Rated speed	115.4r/min	Rated voltage	6kV
Design head	9.5		
Single-unit Discharge capacity	191.5m³/s		
The manufacturer of the turbine(s)	Dongfang Electric Co.,Ltd	The manufacturer of the generator(s)	Dongfang Electric Co.,Ltd
Lifetime of the equipment(years)	30	Lifetime of the equipment(years)	30

After commissioning, the project is operating in good conditions. During this monitoring period, the project was also running orderly. No special event occurred which may impact the GHG emission reductions and monitoring.

3.2 Deviations

3.2.1 Methodology Deviations

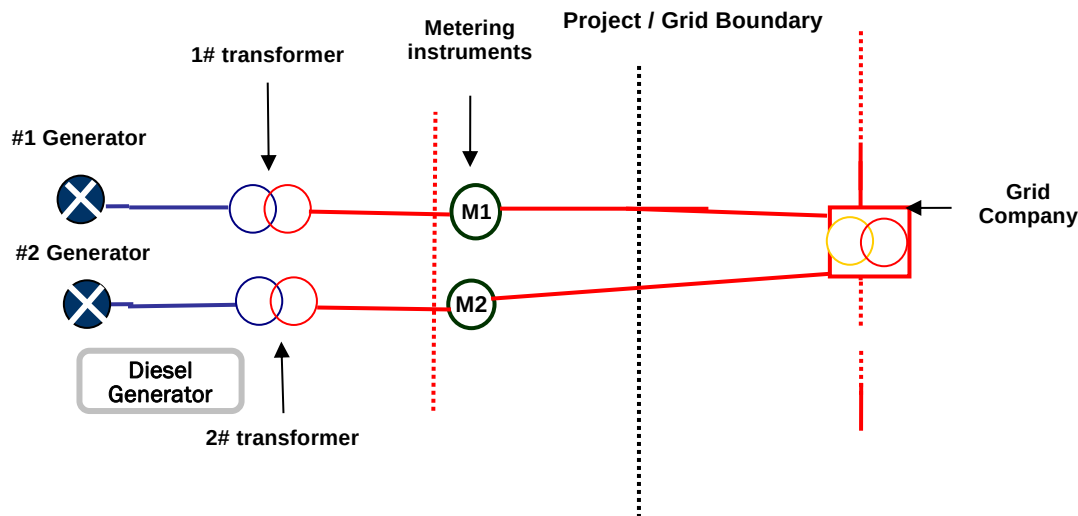
There are no methodology deviations applied during this monitoring period.

3.2.2 Project Description Deviations

As per ACM0002 methodology V10 P11, no leakage emissions are considered. The main emissions potentially giving rise to leakage in the context of electric sector projects are emissions arising due to activities such as power plant construction and upstream emissions from fossil fuel use (e.g. extraction, processing, transport). These emissions sources are neglected. The project is a hydropower project, which does not involve in the leakage emission. However, the registered Project description indicated that the project with a diesel generator used for the purpose of commissioning/maintenance or emergencies during the project implementation stage with leakage emission, which is not consistent with the methodology and a project description deviation is raised, and removed the leakage emission of diesel generator in the leakage emission in the registered PD.

The emission from the diesel generator of the project is within the project boundary, which is perceived as project emission, as per ACM0002 methodology V10 P6, the project emission of the diesel generator of the project could be considered as a minor emission source, that is to say the project emission involved cannot be perceived as significant, therefore the project emission from diesel generator of the project could be neglected.

As per the Power Purchase Agreements (PPAs) and ETNs during this monitoring period of the project, the location of the monitoring meter(s) is showed in the following diagram.



— 110 kV line — 6.3 kV line



Hydro turbine

M1 and M2 are bidirectional meter(s) that measures the power supplied to the grid and the power received from the grid. The measurement results of M1 and M2 are logged manually at 24:00 each day by the plant operation staff and the grid company. The meters of M1(M1_{Ex, M1} and M1_{Im, M1}) and M2(M2_{Ex, M2} and M2_{Im, M2}) are bidirectional and monitored continuously, the grid company and the plant operation staff read and record the meters at the 24:00 of the 27th of each month separately.

Calibrations of meters have been carried out by qualified third party annually in accordance with the revised Monitoring Plan.

The description of the monitoring plan in page 30 of the registered Project description is inconsistent with the implementation of the project, therefore a deviation is requested.

The two deviations of the project as mentioned above, which have no impacts on the applicability of the methodology to the project, the additionality of the project or the appropriateness of the baseline scenario to the project during this monitoring period.

3.3 Grouped Projects

The project is not a group project.

3.4 Baseline Reassessment

Did the project undergo baseline reassessment during the monitoring period?

☐ Yes

☒ No

4 DATA AND PARAMETERS

4.1 Data and Parameters Available at Validation

Data / Parameter	EF _{grid.CM,y}
Data unit	tCO ₂ e/MWh
Description	The baseline combined margin CO ₂ emission factor for grid connected power generation in year y
Source of data	Registered PD
Value applied	0.8888

Justification of choice of data or description of measurement methods and procedures applied	As per the requirements of “Tool to calculate the emission factor for an electricity system” (version 01.1)
Purpose of data	The data is used for calculating the baseline emission
Comments	-

4.2 Data and Parameters Monitored

Data / Parameter	Electricity Exports (EG _{Ex})		
Data unit	MWh		
Description	Annual electricity supplied to the ECPG by the project		
Source of data	Directly measured		
Description of measurement methods and procedures to be applied	Electricity will be measured by meters, and recorded monthly. Data measured by meters will be double checked against receipts from the grid company. The meter(s) will be read frequently and jointly by the Project proponent and the grid company. Data will be then be archived either in computer or in paper. Data monitored are to be kept at least for 2 years after the end of the last crediting period.		
Frequency of monitoring/recording	Measuring continuously/ Recording monthly		
Value monitored	634,403.386		
Monitoring equipment	Meter No	M1	M2
	Type	DTZ341	DTZ341
	Serial Number	EP11143408	EP11143397
	Accuracy class	0.5S	
	Calibration frequency	Yearly	
	On site check date and validity	03-November-2022 to 03-November-2023	
	The calibration of the Meter(s) was not carried out as per the registered PD from 28-March-2006 to 27-March-2012 during this monitoring period, until M1 and M2 calibrated on 03-November-2022, the monitoring meters were calibrated and their accuracy are all within the allowed range and the conservative measures have been applied that is with actual electricity exports minus 0.5% due to the calibration delay.		

QA-QC procedures to be applied	The meter(s) will be read frequently and jointly by the Project proponent and the grid company; keeping the receipts of electricity purchase as a hard proof for data quality control. Detailed descriptions also can be founded in monitoring plan.
Purpose of the data	Calculation of baseline emissions
Calculation method	$EG_{Ex} = EG_{Ex, M1} + EG_{Ex, M2}$
Comments	-

Data / Parameter	Electricity Imports (EG_{im})	
Data unit	MWh	
Description	Annual electricity imported to the project from the ECPG	
Source of data	Directly measured	
Description of measurement methods and procedures to be applied	Electricity will be measured by meters, and recorded monthly. Data measured by meters will be double checked against receipts from the grid company. The meter(s) will be read frequently and jointly by the Project proponent and the grid company. Data will be then be archived either in computer or in paper. Data monitored are to be kept at least for 2 years after the end of the last crediting period.	
Frequency of monitoring-recording	Measuring continuously/ Recording monthly	
Value monitored	0	
Monitoring equipment	Meter No	M1
	Type	DTZ341
	Serial Number	EP11143408
	Accuracy class	0.5S
	Calibration frequency	Yearly
	On site check date and validity	03-November-2022 to 03-November-2023
	The calibration of the Meter(s) was not carried out as per the registered PD from 28-March-2006 to 27-March-2012 during this	

	monitoring period, until M1 and M2 calibrated on 03-November-2022, the monitoring meters were calibrated and their accuracy are all within the allowed range, and the conservative measures have been applied that is with actual electricity imports plus 0.5% due to the calibration delay.
QA/QC procedures to be applied	The meter(s) will be read frequently and jointly by the Project proponent and the grid company; keeping the receipts of electricity purchase as a hard proof for data quality control. Detailed descriptions also can be founded in monitoring plan.
Purpose of the data	Calculation of baseline emissions
Calculation method	$EG_{Im} = EG_{im, M1} + EG_{im, M2}$
Comments	-

Data / Parameter	Net Electricity Exports (EG_y)													
Data unit	MWh													
Description	Net electricity supplied to the ECPG by the project in year y.													
Source of data	Subtract electricity imports (EG_{Im}) from electricity exports (EG_{Ex}), $EG_y = EG_{Ex} - EG_{Im}$													
Description of measurement methods and procedures to be applied	Electricity will be measured by meters, and recorded monthly. Data measured by meters will be double checked against receipts from the grid company. The meter(s) will be read frequently and jointly by the Project proponent and the grid company. Data will be then be archived either in computer or in paper. Data monitored are to be kept at least for 2 years after the end of the last crediting period.													
Frequency of monitoring/recording	Measuring continuously/ Recording monthly													
Value monitored	634,403.386													
Monitoring equipment	<table border="1"> <tr> <td>Meter No</td><td>M1</td><td>M2</td></tr> <tr> <td>Type</td><td>DTZ341</td><td>DTZ341</td></tr> <tr> <td>Serial Number</td><td>EP11143408</td><td>EP11143397</td></tr> <tr> <td>Accuracy class</td><td colspan="2">0.5S</td></tr> </table>		Meter No	M1	M2	Type	DTZ341	DTZ341	Serial Number	EP11143408	EP11143397	Accuracy class	0.5S	
Meter No	M1	M2												
Type	DTZ341	DTZ341												
Serial Number	EP11143408	EP11143397												
Accuracy class	0.5S													

	Calibration frequency	Yearly
	On site check date and validity	03-November-2022 to 03-November-2023
	The calibration of the Meter(s) was not carried out as per the registered PD from 28-March-2006 to 27-March-2012 during this monitoring period, until M1 and M2 calibrated on 03-November-2022, the monitoring meters were calibrated and their accuracy are all within the allowed range, and the conservative measures have been applied due to the calibration delay.	
QA/QC procedures to be applied	-	
Purpose of the data	Calculation of baseline emissions	
Calculation method	-	
Comments	-	

Data / Parameter	A _{PJ}	
Data unit	M ²	
Description	Area of reservoir at full reservoir level	
Source of data	Reports by Fujian Provincial Investigation & Design Institute of Water Conservancy & Hydro Power	
Description of measurement methods and procedures to be applied	Yearly measured from topographical surveys, maps, satellite pictures, etc	
Frequency of monitoring/recording	Measured on yearly basis	
Value monitored	Year	A _{PJ} (m ²)
	2006	1,871,000
	2007	1,871,000
	2008	1,871,000

	2009	1,871,000
	2010	1,871,000
	2011	1,871,000
	2012	1,871,000
Monitoring equipment	-	
QA/QC procedures to be applied	-	
Purpose of the data	Calculation of project emissions	
Calculation method	-	
Comments	-	

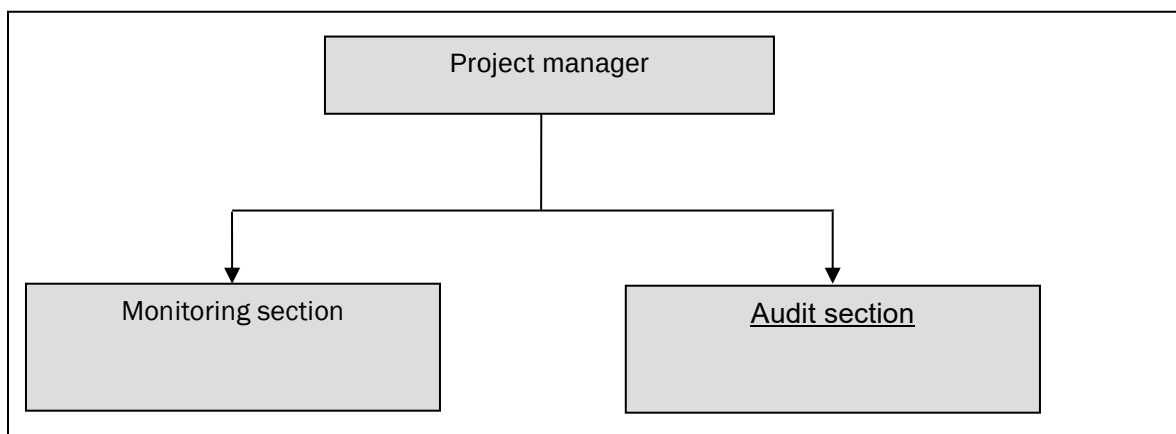
Data / Parameter	Cap _{PI}
Data unit	W
Description	Installed capacity of the hydro power plant after the implementation of the project activity
Source of data	The project site
Description of measurement methods and procedures to be applied	Verified on site
Frequency of monitoring/recording	Yearly monitored based on recognized standards
Value monitored	32,000,000
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of the data	Calculation of project emissions
Calculation method	-
Comments	-

4.3 Monitoring Plan

Organizational Structure

The monitoring of the emission reductions has been carried out according to the scheme below.

The Project proponent will set up a VCS team to be in charge of data collection, supervision, verification and recordation. The project manager will get support from the consultation organization. The structure of the monitoring group is as follows:



The responsibilities of the sections are briefly described as the following:

Project Manager: Receive the summary of monitoring data from Monitoring Section, manage the VCS project jointly with VCS consultant, submit the monitoring report to the DOE and deliver the VCU.

Monitoring Section: Monitor, collect and archive the data according to the Monitoring Plan.

Audit Section: Audit the work of Monitoring Section and execute the QC/QA procedures according to the Monitoring Plan.

Calibration of Meters & Metering

The accuracy of electricity meters is 0.5s. An agreement is to be signed between the Project proponent and the Grid company that defines the metering arrangements and the required quality control procedures to ensure accuracy.

The metering equipment will be properly calibrated and checked annually for accuracy.

Calibration is carried out by Sanming Electricity Bureau with the records being provided to the Project proponent, and these records will be achieved by the Project proponent.

The monitoring system

A bidirectional meter M1 and M2 are located at the high-voltage side of the on-site transformer station, which are measured by the grid company and Project proponent together; it measures the power supplied to the grid ($M1_{Ex, M1}$) and ($M2_{Ex, M2}$) as well as the power received from the grid ($M1_{Im, M1}$) and ($M2_{Im, M2}$). The meters of M1($M1_{Ex, M1}$ and $M1_{Im, M1}$) and M2($M2_{Ex}$ and $M2_{Im}$) are bidirectional and monitored continuously, the grid company and the plant operation staff read and record the meters at the 24:00 of the 27th of each month separately, and the grid company issues ETNs for the electricity exported to the grid and imported from the grid by the readings of two meters of M1 and M2.

Diagram 1 illustrates the positions of the meters and Table 3 shows the information of the monitoring equipment.

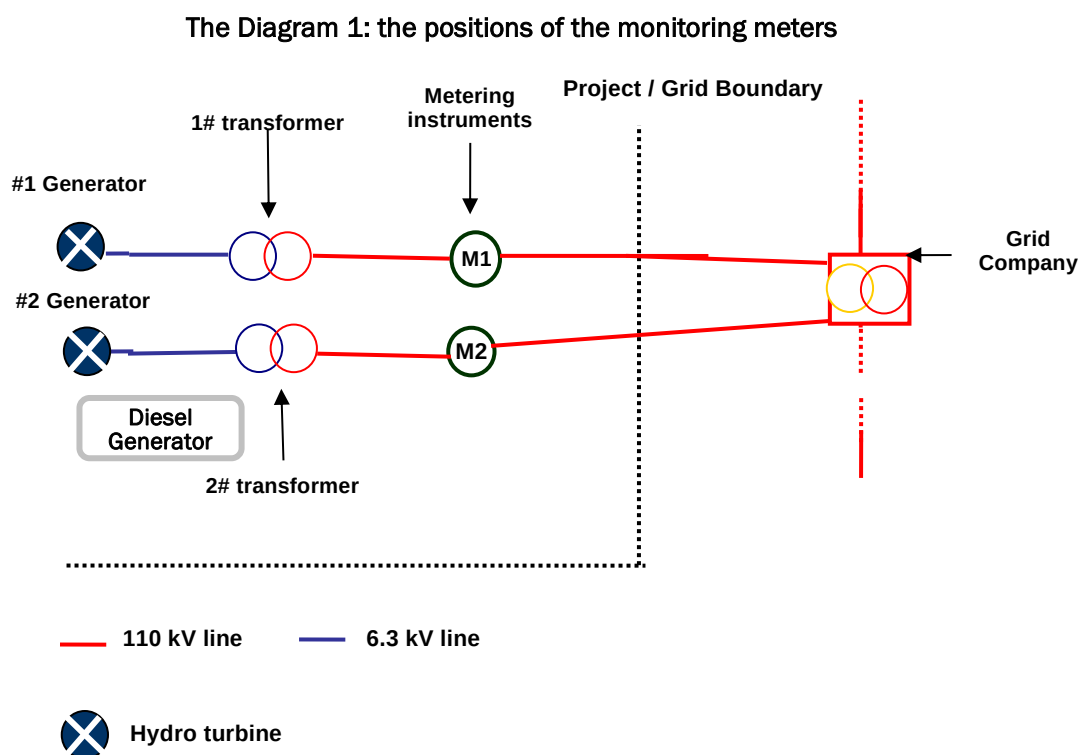


Table 3 The information on monitoring equipment

Meter	Type	Serial Number	Accuracy Class	Calibration Frequency
M1	DTZ341	EP11143408	0.5s	Annually
M2	DTZ341	EP11143397	0.5s	Annually

The Project proponent and the grid company have prepared backup procedures in the Electricity Sales and Purchase Contract to deal with any errors occurred to the meters. In case of meters functioned improperly, or any previous month's reading be inaccurate by more than the allowable error, the grid-connected electricity generated by the project shall be determined by the Project

proponent and the Sanming Electricity Bureau jointly according to other relevant monitoring values. If there is no relevant data available, the Project proponent and the Sanming Electricity Bureau should negotiate about the electricity amount in an appropriate way. No VCS emission reductions will be claimed during such mal-function period of an electrical meter.

The meter reading will be readily accessible for VVB. Calibration tests records will be maintained for verification. There is one diesel generator, which is backup power of flood discharge brake, only started on maintenance reasons to assure the function, or in case of emergency.

Data Calculation Process

To calculate the emission reductions, the annual on-grid electricity supplied to ECPG by the Project (EG_{Ex}) and the annual on-grid electricity purchased from ECPG by the Project (EG_{Im}) are monitored by the meter M1 and M2, and according to the monitoring data, net power supplied to the grid by the project (EG_y) are calculated as $EG_y = EG_{Ex} - EG_{Im} = (M1_{Ex, M1} + M2_{Ex, M2}) - (M1_{Im, M1} + M2_{Im, M2})$.

Quality Assurance and Quality Control

The quality assurance and quality control procedures for recording, maintaining and archiving data shall be improved as part of this VCS project. This is an on-going process that will be ensured through the VCS in terms of the need for verification of the emissions on an annual basis according to this PD and the VCS standard.

Data Management System

The project manager in the VCS team takes the responsibility for keeping all the data collected as part of monitoring archived either in computer or in paper and kept at least for two years after the end of the last crediting period.

Electronical data and documents, will be regularly copied and archived via optical discs or storage tapes, and kept at least for two years after the end of the last crediting period.

Written data and documents, including receipts for double checking of data, will be copied and archived with an explanation of the department or company where the original copy is kept, and kept for at least two years after the end of the last crediting period.

Internal audit and training

All staff involved in the monitoring process will be trained. The training contains VCS knowledge, operational regulations, data monitoring requirements and data management regulations, etc.

A site-specific monitoring manual comprising monitoring procedures will be developed and implemented to ensure a robust data gathering system at the project location. All relevant

monitoring personnel will receive specific training and responsibilities within the monitoring process.

5 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

5.1 Baseline Emissions

According to ACM0002 (Version 10), the emission reduction calculation is carried out as following:

Baseline emission calculation

Baseline emissions are calculated with combined baseline emission factor and the net electricity delivered to the grid by the project. The baseline emission (BE_y) shall be determined as follows:

$$BE_y = (EG_y - EG_{baseline}) \times EF_{grid,CM,y} \quad (1)$$

Where:

BE_y = Baseline emissions in year y (tCO₂e/y);

EG_y = Net electricity supplied by the project activity to the grid (MWh);

EG_{baseline} = Baseline electricity supplied to the grid in the case of modified or retrofit facilities (MWh). For new power plants this value is taken as zero;

EF_{grid, CM, y} = Combined margin CO₂ emission factor for grid connected power generation in year y calculated using the latest version of the “Tool to calculate the emission factor for an electricity system” (tCO₂e/MWh) .

Since the project installs a new grid-connected renewable power plant/unit, EG_{baseline} = 0; therefore, BE_y = EG_y × EF_{grid,CM,y}.

In accordance with the above stated monitoring parameters in section 4.3, for the project activity:

$$EG_y = EG_{Ex} - EG_{im} \quad (2)$$

EG_{Ex} is annual on-grid electricity supplied to ECPG by the project

EG_{im} is annual on-grid electricity purchased from ECPG by the project for the plant operation.

Baseline emissions calculations

The baseline emissions calculations are shown in Table 4 and Appendix III.

The annually and monthly baseline emissions calculations are shown in Table 4 and Appendix 3.

Table 4 Annually Baseline emission calculation

Period	EG _y (MWh)	EF _{grid,CM, y} (tCO ₂ e/MWh)	BE _y (tCO ₂ e)
	A	B	C=Rounddown(A X B)
28-March-2006 to 31-December -2006	98,091.754	0.8888	87,183
01-January-2007 to 31-December -2007	110,536.769	0.8888	98,245
01-January-2008 to 31-December -2008	111,826.879	0.8888	99,391
01-January-2009 to 31-December-2009	79,645.246	0.8888	70,788
01- January -2010 to 31-December -2010	127,256.995	0.8888	113,106
01- January -2011 to 31-December -2011	79,427.578	0.8888	70,595
01-January-2012 to 27-March-2012	27,618.166	0.8888	24,547
In total	634,403.386	-	563,855

5.2 Project Emissions

As per ACM0002 version 10, the project emissions of the project equal to the emissions from the reservoir.

The power density of the project in each year of this monitoring period is listed in Table 5 based on the installed capacity of the hydropower plant and the surface area of the reservoir.

Table 5. The power density of the project in this monitoring period

Year	Cap _{PJ} (W)	Cap _{BL} (W)	A _{PJ} (m ²)	A _{BL} (m ²)	Power Density (W/m ²)
	A	B	C	D	E=(A-B)/(C-D)
2006	32,000,000	0	1,871,000	0	17.10
2007	32,000,000	0	1,871,000	0	17.10
2008	32,000,000	0	1,871,000	0	17.10
2009	32,000,000	0	1,871,000	0	17.10
2010	32,000,000	0	1,871,000	0	17.10
2011	32,000,000	0	1,871,000	0	17.10
2012	32,000,000	0	1,871,000	0	17.10

The power density in each year of this monitoring period is above the threshold of 10 W/m², as per the methodology ACM0002 version 10, the emissions from the reservoir are neglected.

Therefore, the project emission is 0.

5.3 Leakage Emissions

No leakage in the project according to methodology, i.e. LEy = 0

5.4 GHG Emission Reductions and Carbon Dioxide Removals

Vintage period	Baseline emissions (tCO ₂ e)	Project emissions (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Reduction VCUs (tCO ₂ e)	Removal VCUs (tCO ₂ e)	Total VCUs (tCO ₂ e)
28-March-2006 to 31- December -2006	87,183	0	0	87,183	0	87,183
01-January-2007 to 31- December -2007	98,245	0	0	98,245	0	98,245
01-January-2008 to 31- December -2008	99,391	0	0	99,391	0	99,391
01-January-2009 to 31- December -2009	70,788	0	0	70,788	0	70,788
01- January - 2010 to 31- December -2010	113,106	0	0	113,106	0	113,106
01- January - 2011 to 31- December -2011	70,595	0	0	70,595	0	70,595
01- January - 2012 to 27- March-2012	24,547	0	0	24,547	0	24,547
Total	563,855	0	0	563,855	0	563,855

Vintage period	Ex-ante estimated reductions/ removals	Achieved reductions/ removals	Percent difference	Explanation for the difference
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28-March-2006 to 31-December-2006	87,729 ⁷	87,183	-0.62%	Normal Variation
01-January-2007 to 31- December - 2007	98,629	98,245	-0.39%	Normal Variation
01-January-2008 to 31- December - 2008	99,922	99,391	-0.53%	Normal Variation
01-January-2009 to 31-December - 2009	106,906	70,788	-33.78%	The water flow in 2009 is much lower than the estimated average water flow ⁸ .
01-January -2010 to 31-December - 2010	106,906	113,106	5.80%	The water flow in 2010 is higher than the estimated average water flow but within 10% sensitivity analysis range.
001-January -2011 to 31- December - 2011	106,906	70,595	-33.97%	The water flow in 2011 is much lower than the estimated average water flow ⁹ .
001-January -2012 to 27-March-2012	25,481	24,547	-3.67%	The water flow in the first quarter of 2012 is lower than the estimated average water flow, as it is the quarter of low water flow in a year but within 10% sensitivity analysis range.
Total	636,520	563,855	-11.42%	The water flow on average during this monitoring period is less than that of the estimated average water flow, and the actual ERs is 11.42% less than the estimated total ERs based on the Ex-ante estimated emission reductions

⁷From 28-March-2006 to 27-December-2008, the ERs in the registered PD P2 is used for vintage period calculation, the other vintage period is annual ex-ante ER of 106,906 tCO₂e/y.

⁸ According to 2009 Fujian Provincial Environmental Status communiqué, the water flow is lower water flow year, therefore the power generation is much lower than the estimated average.

⁹ According to 2011 Fujian Provincial Environmental Status communiqué, the water flow is lower water flow year, therefore the power generation is much lower than the estimated average.

		per year in the registered PD, which is reasonable in the project scenario.
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APPENDIX 1: COMMERCIALY SENSITIVE INFORMATION

No commercially sensitive information.

APPENDIX II: < EVIDENCES OF SUSTAINABLE DEVELOPMENT GOAL >

Evidence for SDG 8.5

2006 年福建将乐大言 32MW 水电项目年工资单
Staff name-list and salary of the project in 2006

序号 NO	姓名 Name	性别 Gender	年工资 Annual salary
1.	林建森	男	38760
2.	郑立文	男	38372.4
3.	戴洪铭	男	37988.68
4.	饶海盛	女	37608.79
5.	廖应灿	男	37232.7
6.	黄长丽	女	36860.37
7.	刘爱英	女	36491.77
8.	林鹏	男	36126.85
9.	杨国强	男	35765.58
10.	谢珊英	女	35407.93
11.	廖芳吉	男	35053.85
12.	陈勇	女	34703.31
13.	孙玉兵	男	34356.28
14.	龚盱刚	男	34012.71
15.	陈琳	女	33672.59
16.	林金新	女	33335.86
17.	林诗卉	女	33002.5
18.	曾至冬	女	32672.48
19.	朱惠山	男	32345.75
20.	黄子博	男	32022.3
21.	陈满华	女	31702.07
22.	郑贝娜	女	31385.05
23.	刘小华	女	31071.2
24.	彭立勇	男	30760.49
25.	江秀羽	女	30452.88
26.	李志辉	男	30148.36
27.	乐南海	男	29846.87
28.	江青应	女	29548.4
29.	陈利涛	男	29252.92
30.	江宁	女	28960.39
31.	叶箭星	女	28670.79
32.	邹德仁	男	28384.08
33.	张国强	男	28100.24
34.	邹小萍	女	27819.24
平均工资			32996.93

2007 年福建将乐大官 32MW 水电项目年工资单

Staff name-list and salary of the project in 2007

序号 NO	姓名 Name	性别 Gender	年工资 Annual salary
1.	林建森	男	40560
2.	郑立文	男	40154.4
3.	戴洪铭	男	39752.86
4.	饶海盛	女	39355.33
5.	廖应灿	男	38961.77
6.	黄长丽	女	38572.16
7.	刘爱英	女	38186.43
8.	林鹏	男	37804.57
9.	杨国强	男	37426.52
10.	谢珊英	女	37052.26
11.	廖芳吉	男	36681.74
12.	陈勇	女	36314.92
13.	孙玉兵	男	35951.77
14.	龚盱刚	男	35592.25
15.	陈琳	女	35236.33
16.	林金新	女	34883.97
17.	林诗卉	女	34535.13
18.	曾至冬	女	34189.78
19.	朱惠山	男	33847.88
20.	黄子博	男	33509.4
21.	陈潇华	女	33174.21
22.	郑贝娜	女	32842.56
23.	刘小华	女	32514.14
24.	彭立勇	男	32189
25.	江秀羽	女	31867.11
26.	李志辉	男	31548.43
27.	乐南海	男	31232.95
28.	江青应	女	30920.62
29.	陈利涛	男	30611.41
30.	江宁	女	30305.3
31.	叶箭星	女	30002.25
32.	邹德仁	男	29702.22
33.	张国强	男	29405.2
34.	邹小萍	女	29111.15
平均工资			34529.3

2008 年福建将乐大官 32MW 水电项目年工资单

Staff name-list and salary of the project in 2008

序号 NO	姓名 Name	性别 Gender	年工资 Annual salary
1.	林建森	男	42960
2.	郑立文	男	42530.4
3.	戴洪铭	男	42105.1
4.	饶海盛	女	41684.05
5.	廖应灿	男	41267.2
6.	黄长丽	女	40854.53
7.	刘爱英	女	40445.99
8.	林鹏	男	40041.53
9.	杨国强	男	39641.11
10.	谢珊英	女	39244.7
11.	廖芳吉	男	38852.25
12.	陈勇	女	38463.73
13.	孙玉兵	男	38079.09
14.	龚盱刚	男	37698.3
15.	陈琳	女	37321.32
16.	林金新	女	36948.11
17.	林诗卉	女	36578.63
18.	曾至冬	女	36212.84
19.	朱惠山	男	35850.71
20.	黄子博	男	35492.2
21.	陈潇华	女	35137.28
22.	郑贝娜	女	34785.91
23.	刘小华	女	34438.05
24.	彭立勇	男	34093.67
25.	江秀羽	女	33752.73
26.	李志辉	男	33415.21
27.	乐南海	男	33081.05
28.	江青应	女	32750.24
29.	陈利涛	男	32422.74
30.	江宁	女	32098.51
31.	叶箭星	女	31777.53
32.	邹德仁	男	31459.75
33.	张国强	男	31145.16
34.	邹小萍	女	30833.7
平均工资			36572.45

2009 年福建将乐大官 32MW 水电项目年工资单
Staff name-list and salary of the project in 2009

序号 NO	姓名 Name	性别 Gender	年工资 Annual salary
1.	林建森	男	44160
2.	郑立文	男	43718.4
3.	戴洪铭	男	43281.22
4.	饶海盛	女	42848.4
5.	廖应灿	男	42419.92
6.	黄长丽	女	41995.72
7.	刘爱英	女	41575.76
8.	林鹏	男	41160.01
9.	杨国强	男	40748.41
10.	谢珊英	女	40340.92
11.	廖芳吉	男	39937.51
12.	陈勇	女	39538.14
13.	孙玉兵	男	39142.76
14.	龚盱刚	男	38751.33
15.	陈琳	女	38363.82
16.	林金新	女	37980.18
17.	林诗卉	女	37600.38
18.	曾至冬	女	37224.37
19.	朱惠山	男	36852.13
20.	黄子博	男	36483.61
21.	陈蒲华	女	36118.77
22.	郑贝娜	女	35757.58
23.	刘小华	女	35400.01
24.	彭立勇	男	35046.01
25.	江秀羽	女	34695.55
26.	李志辉	男	34348.59
27.	乐南海	男	34005.11
28.	江青应	女	33665.05
29.	陈利涛	男	33328.4
30.	江宁	女	32995.12
31.	叶箭星	女	32665.17
32.	邹德仁	男	32338.52
33.	张国强	男	32015.13
34.	邹小萍	女	31694.98
平均工资			37594.03

2010 年福建将乐大官 32MW 水电项目年工资单
Staff name-list and salary of the project in 2010

序号 NO	姓名 Name	性别 Gender	年工资 Annual salary
1.	林建森	男	45588
2.	郑立文	男	45132.12
3.	戴洪铭	男	44680.8
4.	饶海盛	女	44233.99
5.	廖应灿	男	43791.65
6.	黄长丽	女	43353.73
7.	刘爱英	女	42920.2
8.	林鹏	男	42491
9.	杨国强	男	42066.09
10.	谢珊英	女	41645.42
11.	廖芳吉	男	41228.97
12.	陈勇	女	40816.68
13.	孙玉兵	男	40408.51
14.	龚时刚	男	40004.43
15.	陈琳	女	39604.38
16.	林金新	女	39208.34
17.	林诗卉	女	38816.26
18.	曾至冬	女	38428.09
19.	朱惠山	男	38043.81
20.	黄子博	男	37663.38
21.	陈潇华	女	37286.74
22.	郑贝娜	女	36913.97
23.	刘小华	女	36544.74
24.	彭立勇	男	36179.29
25.	江秀羽	女	35817.5
26.	李志辉	男	35459.22
27.	乐南海	男	35104.73
28.	江青应	女	34753.68
29.	陈利涛	男	34406.14
30.	江宁	女	34062.08
31.	叶箭星	女	33721.46
32.	邹德仁	男	33384.25
33.	张国强	男	33050.4
34.	邹小萍	女	32719.9
平均工资			38809.7

2011 年福建将乐大言 32MW 水电项目年工资单

Staff name-list and salary of the project in 2011

序号 NO	姓名 Name	性别 Gender	年工资 Annual salary
1.	林建森	男	49000
2.	郑立文	男	47520
3.	戴洪铭	男	47044.8
4.	饶海盛	女	46574.35
5.	廖应灿	男	46108.61
6.	黄长丽	女	45647.52
7.	刘爱英	女	45191.05
8.	林鹏	男	44739.14
9.	杨国强	男	44291.75
10.	谢珊英	女	43848.83
11.	廖芳吉	男	43410.34
12.	陈勇	女	42976.24
13.	孙玉兵	男	42546.47
14.	龚盱刚	男	42121.01
15.	陈琳	女	41699.8
16.	林金新	女	41282.8
17.	林诗卉	女	40869.97
18.	曾至冬	女	40461.27
19.	朱惠山	男	40056.66
20.	黄子博	男	39656.09
21.	陈潇华	女	39259.53
22.	郑贝娜	女	38866.94
23.	刘小华	女	38478.27
24.	彭立勇	男	38093.49
25.	江秀羽	女	37712.55
26.	李志辉	男	37335.43
27.	乐南海	男	36962.07
28.	江青应	女	36592.45
29.	陈利涛	男	36226.53
30.	江宁	女	35864.26
31.	叶箭星	女	35505.62
32.	邹德仁	男	35150.56
33.	张国强	男	34799.06
34.	邹小萍	女	34451.07
平均工资			40863.07

2012 年福建将乐大官 32MW 水电项目年工资单

Staff name-list and salary of the project in 2012

序号 NO	姓名 Name	性别 Gender	年工资 Annual salary
1.	林建森	男	49224
2.	郑立文	男	48731.76
3.	戴洪铭	男	48244.44
4.	饶海盛	女	47762
5.	廖应灿	男	47284.38
6.	黄长丽	女	46811.53
7.	刘爱英	女	46343.42
8.	林鹏	男	45879.98
9.	杨国强	男	45421.18
10.	谢珊英	女	44966.97
11.	廖芳吉	男	44517.3
12.	陈勇	女	44072.13
13.	孙玉兵	男	43631.41
14.	龚盱刚	男	43195.09
15.	陈琳	女	42763.14
16.	林金新	女	42335.51
17.	林诗卉	女	41912.16
18.	曾至冬	女	41493.04
19.	朱惠山	男	41078.11
20.	黄子博	男	40667.32
21.	陈潇华	女	40260.65
22.	郑贝娜	女	39859.04
23.	刘小华	女	39459.46
24.	彭立勇	男	39064.87
25.	江秀羽	女	38674.22
26.	李志辉	男	38287.48
27.	乐南海	男	37904.6
28.	江青应	女	37525.56
29.	陈利涛	男	37150.3
30.	江宁	女	36778.8
31.	叶箭星	女	36411.01
32.	邹德仁	男	36046.9
33.	张国强	男	35686.43
34.	邹小萍	女	35329.57
平均工资			41905.08

Evidence for SDG7.2 and 13.0

The electricity supplied during this monitoring period from 28-March-2006 to 27-March-2012 is 634,403.386MWh. The total emission reductions claimed in this monitoring period are 563,855tCO₂e. and the emission reduction generated from the project during this monitoring period is 563,855tCO₂e as mentioned in section 1.12 table 1 above, which is calculated from meter(s) readings and ETNs provided by the Project proponent, which would be verified by VVB.

APPENDIX III: < MONTHLY BASELINE EMISSIONS CALCULATIONS >

Monitoring Period	Meter reading				ETNs		Cross check between EG_{Ex} without Calibration (Electricity supplied by the project activity to ECPG) and ETNs Electricity supplied by the project activity to ECPG) (MWh)	Max(EG_{Im} without Calibration, y) (Electricity consumption from the grid, ETNs Electricity consumption from the grid) (MWh)	EG_y (MWh)	$EF_{grid,CM, y}$ (tCO ₂ e/MWh)	Baseline emissions (tCO ₂ e)
	EG_{Ex} (Electricity supplied by the project activity to ECPG) MWh	EG_{Ex} without calibration (Electricity supplied by the project activity to ECPG) $EG_{Ex}*(1-0.5\%)$ MWh	EG_{Im} (Electricity consumption from the grid) MWh	EG_{Im} without calibration (Electricity consumption from the grid) $EG_{Im}*(1+0.5\%)$ MWh	(Electricity exported to the Grid) (MWh)	(Electricity consumption from the Grid) (MWh)					
	A	B	C	D	E	F					
28-March-2006 to 27-April-2006	12,248.410	12,187.168	0	0	12,248.410	0	12,187.168	0.000	12,187.168	0.8888	10,832
28-April-2006 to 27-May-2006	14,056.210	13,985.929	0	0	14,056.210	0	13,985.929	0.000	13,985.929	0.8888	12,431
28-May-2006 to 27-June-2006	17,585.315	17,497.388	0	0	17,585.315	0	17,497.388	0.000	17,497.388	0.8888	15,552
28-June-2006 to 27-July-2006	14,514.141	14,441.570	0	0	14,514.141	0	14,441.570	0.000	14,441.570	0.8888	12,836
28-July-2006 to 27-August-2006	11,983.654	11,923.736	0	0	11,983.654	0	11,923.736	0.000	11,923.736	0.8888	10,598
28-August-2006 to 27-September-2006	8,616.409	8,573.327	0	0	8,616.409	0	8,573.327	0.000	8,573.327	0.8888	7,620
28-September-2006 to 27-October-2006	6,487.078	6,454.643	0	0	6,487.078	0	6,454.643	0.000	6,454.643	0.8888	5,737
28-October-2006 to 27-November-2006	5,378.397	5,351.505	0	0	5,378.397	0	5,351.505	0.000	5,351.505	0.8888	4,756
28-November-2006 to 27-December-2006	6,791.943	6,757.983	0	0	6,791.943	0	6,757.983	0.000	6,757.983	0.8888	6,006

28-December-2006 to 31-December-2006	923.120	918.504	0	0	923.120	0	918.504	0.000	918.504	0.8888	816
Sub-total of 2006	98,584.677	98,091.754	0	0	98,584.677	0	98,091.754	0	98,091.754	-	87,183¹⁰
01-January-2007 to 27-January-2007	6,231.060	6,199.905	0	0	6,231.060	0	6,199.905	0.000	6,199.905	0.8888	5,510
28-January-2007 to 27-February-2007	13,028.223	12,963.082	0	0	13,028.223	0	12,963.082	0.000	12,963.082	0.8888	11,522
28-February-2007 to 27-March-2007	13,534.066	13,466.396	0	0	13,534.066	0	13,466.396	0.000	13,466.396	0.8888	11,969
28-March-2007 to 27-April-2007	12,257.981	12,196.691	0	0	12,257.981	0	12,196.691	0.000	12,196.691	0.8888	10,840
28-April-2007 to 27-May-2007	8,922.476	8,877.864	0	0	8,922.476	0	8,877.864	0.000	8,877.864	0.8888	7,891
28-May-2007 to 27-June-2007	12,067.697	12,007.359	0	0	12,067.697	0	12,007.359	0.000	12,007.359	0.8888	10,672
28-June-2007 to 27-July-2007	9,382.312	9,335.400	0	0	9,382.312	0	9,335.400	0.000	9,335.400	0.8888	8,297
28-July-2007 to 27-August-2007	7,992.079	7,952.119	0	0	7,992.079	0	7,952.119	0.000	7,952.119	0.8888	7,068
28-August-2007 to 27-September-2007	12,833.064	12,768.899	0	0	12,833.064	0	12,768.899	0.000	12,768.899	0.8888	11,349
28-September-2007 to 27-October-2007	8,687.254	8,643.818	0	0	8,687.254	0	8,643.818	0.000	8,643.818	0.8888	7,683
28-October-2007 to 27-November-2007	3,150.540	3,134.787	0	0	3,150.540	0	3,134.787	0.000	3,134.787	0.8888	2,786
28-November-2007 to 27-December-2007	2,678.601	2,665.208	0	0	2,678.601	0	2,665.208	0.000	2,665.208	0.8888	2,369
28-December-2007 to 31-December-2007	326.877	325.243	0	0	326.877	0	325.243	0.000	325.243	0.8888	289

¹⁰ Only the annually baseline emissions round-down, the monthly baseline emissions do not round-down, hereafter of the following year are the same.

Sub-total of 2007	111,092.230	110,536.769	0	0	111,092.230	0	110,536.769	0	110,536.769	-	98,245
01-January-2008 to 27-January-2008	2,206.416	2,195.384	0	0	2,206.416	0	2,195.384	0.000	2,195.384	0.8888	1,951
28-January-2008 to 27-February-2008	7,075.983	7,040.603	0	0	7,075.983	0	7,040.603	0.000	7,040.603	0.8888	6,258
28-February-2008 to 27-March-2008	6,404.621	6,372.598	0	0	6,404.621	0	6,372.598	0.000	6,372.598	0.8888	5,664
28-March-2008 to 27-April-2008	14,675.304	14,601.927	0	0	14,675.304	0	14,601.927	0.000	14,601.927	0.8888	12,978
28-April-2008 to 27-May-2008	13,698.976	13,630.481	0	0	13,698.976	0	13,630.481	0.000	13,630.481	0.8888	12,115
28-May-2008 to 27-June-2008	11,914.820	11,855.246	0	0	11,914.820	0	11,855.246	0.000	11,855.246	0.8888	10,537
28-June-2008 to 27-July-2008	14,299.694	14,228.196	0	0	14,299.694	0	14,228.196	0.000	14,228.196	0.8888	12,646
28-July-2008 to 27-August-2008	13,495.259	13,427.783	0	0	13,495.259	0	13,427.783	0.000	13,427.783	0.8888	11,935
28-August-2008 to 27-September-2008	8,307.810	8,266.271	0	0	8,307.810	0	8,266.271	0.000	8,266.271	0.8888	7,347
28-September-2008 to 27-October-2008	5,127.911	5,102.271	0	0	5,127.911	0	5,102.271	0.000	5,102.271	0.8888	4,535
28-October-2008 to 27-November-2008	6,394.968	6,362.993	0	0	6,394.968	0	6,362.993	0.000	6,362.993	0.8888	5,655
28-November-2008 to 27-December-2008	8,258.882	8,217.588	0	0	8,258.882	0	8,217.588	0.000	8,217.588	0.8888	7,304
28-December-2008 to 31-December-2008	528.179	525.538	0	0	528.179	0	525.538	0.000	525.538	0.8888	467
Sub-total of 2008	112,388.823	111,826.879	0	0	112,388.823	0	111,826.879	0	111,826.879	-	99,391
01-January-2009 to 27-January-2009	3,565.209	3,547.383	0	0	3,565.209	0	3,547.383	0.000	3,547.383	0.8888	3,153
28-January-2009 to 27-February-2009	8,517.191	8,474.605	0	0	8,517.191	0	8,474.605	0.000	8,474.605	0.8888	7,532

28-February-2009 to 27-March-2009	2,987.578	2,972.640	0	0	2,987.578	0	2,972.640	0.000	2,972.640	0.8888	2,642
28-March-2009 to 27-April-2009	9,837.276	9,788.090	0	0	9,837.276	0	9,788.090	0.000	9,788.090	0.8888	8,700
28-April-2009 to 27-May-2009	5,866.150	5,836.819	0	0	5,866.150	0	5,836.819	0.000	5,836.819	0.8888	5,188
28-May-2009 to 27-June-2009	9,837.276	9,788.090	0	0	9,837.276	0	9,788.090	0.000	9,788.090	0.8888	8,700
28-June-2009 to 27-July-2009	5,866.150	5,836.819	0	0	5,866.150	0	5,836.819	0.000	5,836.819	0.8888	5,188
28-July-2009 to 27-August-2009	5,762.810	5,733.996	0	0	5,762.810	0	5,733.996	0.000	5,733.996	0.8888	5,096
28-August-2009 to 27-September-2009	2,282.699	2,271.286	0	0	2,282.699	0	2,271.286	0.000	2,271.286	0.8888	2,019
28-September-2009 to 27-October-2009	6,485.667	6,453.239	0	0	6,485.667	0	6,453.239	0.000	6,453.239	0.8888	5,736
28-October-2009 to 27-November-2009	7,985.281	7,945.355	0	0	7,985.281	0	7,945.355	0.000	7,945.355	0.8888	7,062
28-November-2009 to 27-December-2009	9,865.129	9,815.803	0	0	9,865.129	0	9,815.803	0.000	9,815.803	0.8888	8,724
28-December-2009 to 31-December-2009	1,187.057	1,181.122	0	0	1,187.057	0	1,181.122	0.000	1,181.122	0.8888	1,050
Sub-total of 2009	80,045.473	79,645.246	0	0	80,045.473	0	79,645.246	0	79,645.246	-	70,788
01-January-2010 to 27-January-2010	8,012.638	7,972.575	0	0	8,012.638	0	7,972.575	0.000	7,972.575	0.8888	7,086
28-January-2010 to 27-February-2010	8,268.972	8,227.627	0	0	8,268.972	0	8,227.627	0.000	8,227.627	0.8888	7,313
28-February-2010 to 27-March-2010	8,115.272	8,074.696	0	0	8,115.272	0	8,074.696	0.000	8,074.696	0.8888	7,177
28-March-2010 to 27-April-2010	15,310.568	15,234.015	0	0	15,310.568	0	15,234.015	0.000	15,234.015	0.8888	13,540
28-April-2010 to 27-May-2010	17,947.391	17,857.654	0	0	17,947.391	0	17,857.654	0.000	17,857.654	0.8888	15,872
28-May-2010 to 27-June-2010	15,759.137	15,680.341	0	0	15,759.137	0	15,680.341	0.000	15,680.341	0.8888	13,937

28-June-2010 to 27-July-2010	17,886.728	17,797.294	0	0	17,886.728	0	17,797.294	0.000	17,797.294	0.8888	15,818
28-July-2010 to 27-August-2010	13,456.874	13,389.590	0	0	13,456.874	0	13,389.590	0.000	13,389.590	0.8888	11,901
28-August-2010 to 27-September-2010	11,498.951	11,441.456	0	0	11,498.951	0	11,441.456	0.000	11,441.456	0.8888	10,169
28-September-2010 to 27-October-2010	1,935.489	1,925.812	0	0	1,935.489	0	1,925.812	0.000	1,925.812	0.8888	1,712
28-October-2010 to 27-November-2010	4,267.728	4,246.389	0	0	4,267.728	0	4,246.389	0.000	4,246.389	0.8888	3,774
28-November-2010 to 27-December-2010	4,107.166	4,086.630	0	0	4,107.166	0	4,086.630	0.000	4,086.630	0.8888	3,632
28-December-2010 to 31-December-2010	1,329.563	1,322.915	0	0	1,329.563	0	1,322.915	0.000	1,322.915	0.8888	1,176
Sub-total of 2010	127,896.477	127,256.995	0	0	127,896.477	0	127,256.995	0	127,256.995	-	113,106
01-January-2011 to 27-January-2011	10,304.116	10,252.595	0	0	10,304.116	0	10,252.595	0.000	10,252.595	0.8888	9,113
28-January-2011 to 27-February-2011	11,457.195	11,399.909	0	0	11,457.195	0	11,399.909	0.000	11,399.909	0.8888	10,132
28-February-2011 to 27-March-2011	4,885.431	4,861.004	0	0	4,885.431	0	4,861.004	0.000	4,861.004	0.8888	4,320
28-March-2011 to 27-Apr-2011	5,113.970	5,088.400	0	0	5,113.970	0	5,088.400	0.000	5,088.400	0.8888	4,523
28-April-2011 to 27-May-2011	7,939.853	7,900.154	0	0	7,939.853	0	7,900.154	0.000	7,900.154	0.8888	7,022
28-May-2011 to 27-June-2011	4,408.006	4,385.966	0	0	4,408.006	0	4,385.966	0.000	4,385.966	0.8888	3,898
28-June-2011 to 27-July-2011	4,803.344	4,779.327	0	0	4,803.344	0	4,779.327	0.000	4,779.327	0.8888	4,248
28-July-2011 to 27-August-2011	5,487.977	5,460.537	0	0	5,487.977	0	5,460.537	0.000	5,460.537	0.8888	4,853
28-August-2011 to 27-September-2011	8,308.911	8,267.366	0	0	8,308.911	0	8,267.366	0.000	8,267.366	0.8888	7,348

28-September-2011 to 27-October-2011	4,640.517	4,617.314	0	0	4,640.517	0	4,617.314	0.000	4,617.314	0.8888	4,104
28-October-2011 to 27-November-2011	4,784.057	4,760.137	0	0	4,784.057	0	4,760.137	0.000	4,760.137	0.8888	4,231
28-November-2011 to 27-December-2011	7,003.526	6,968.508	0	0	7,003.526	0	6,968.508	0.000	6,968.508	0.8888	6,194
28-December-2011 to 31-December-2011	689.809	686.360	0	0	689.809	0	686.360	0.000	686.360	0.8888	610
Sub-total of 2011	79,826.712	79,427.578	0	0	79,826.712	0	79,427.578	0	79,427.578	-	70,595
01-January-2012 to 27-January-2012	5,345.995	5,319.265	0	0	5,345.995	0	5,319.265	0.000	5,319.265	0.8888	4,728
28-January-2012 to 27-February-2012	8,853.890	8,809.621	0	0	8,853.890	0	8,809.621	0.000	8,809.621	0.8888	7,830
28-February-2012 to 27-March-2012	13,557.066	13,489.281	0	0	13,557.066	0	13,489.281	0.000	13,489.281	0.8888	11,989
Sub-total of 2012	27,756.951	27,618.166	0.000	0.000	27,756.951	0.000	27,618.166	0.000	27,618.166	-	24,547
In total	637,591.343	634,403.386	0.000	0.000	637,591.343	0.000	634,403.386	0	634,403.386	-	563,855